

Automated Control System for Disposable Cup Manufacturing

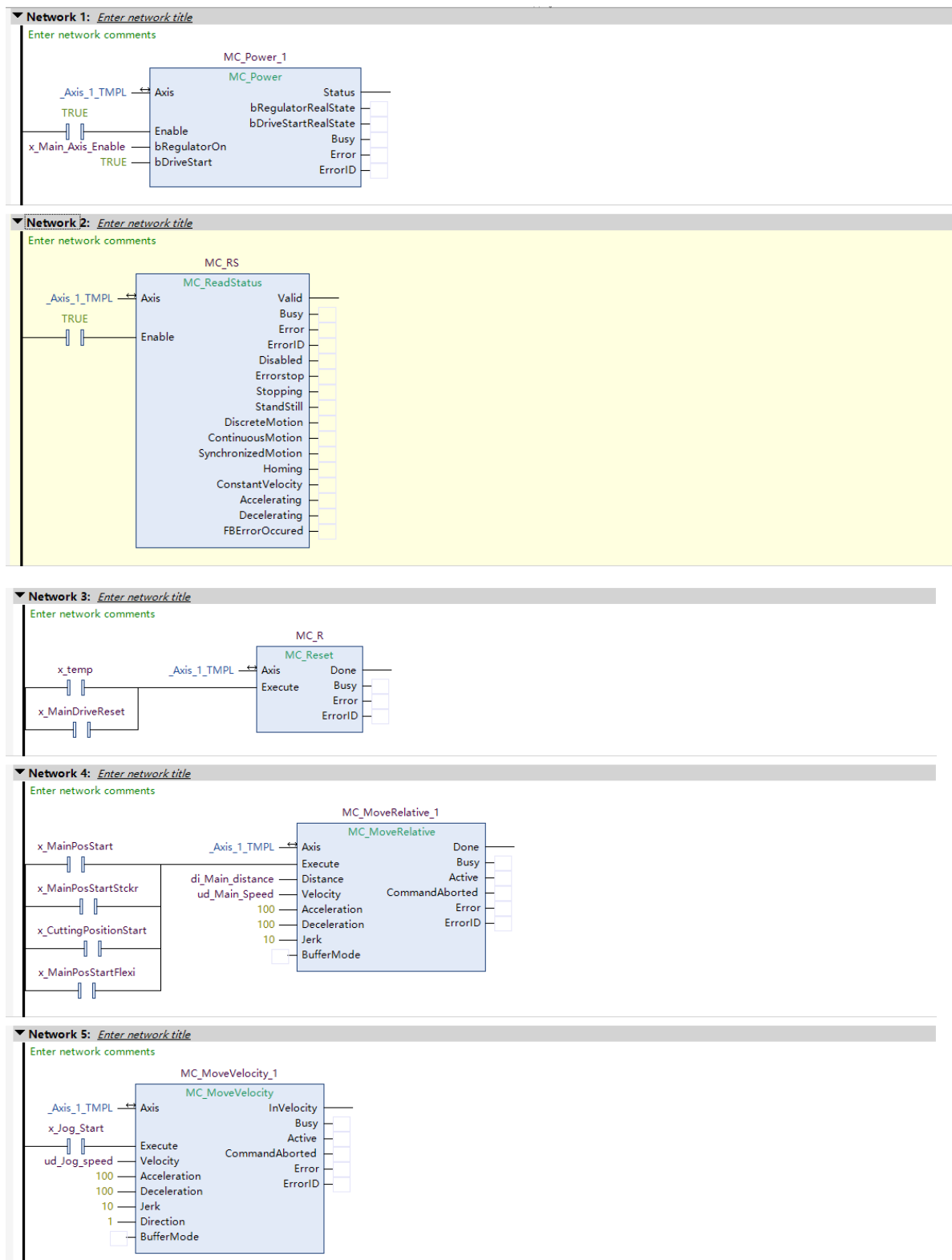
Polypropylene shards are used as a raw material for this machine. It is commercially bought in large quantities.

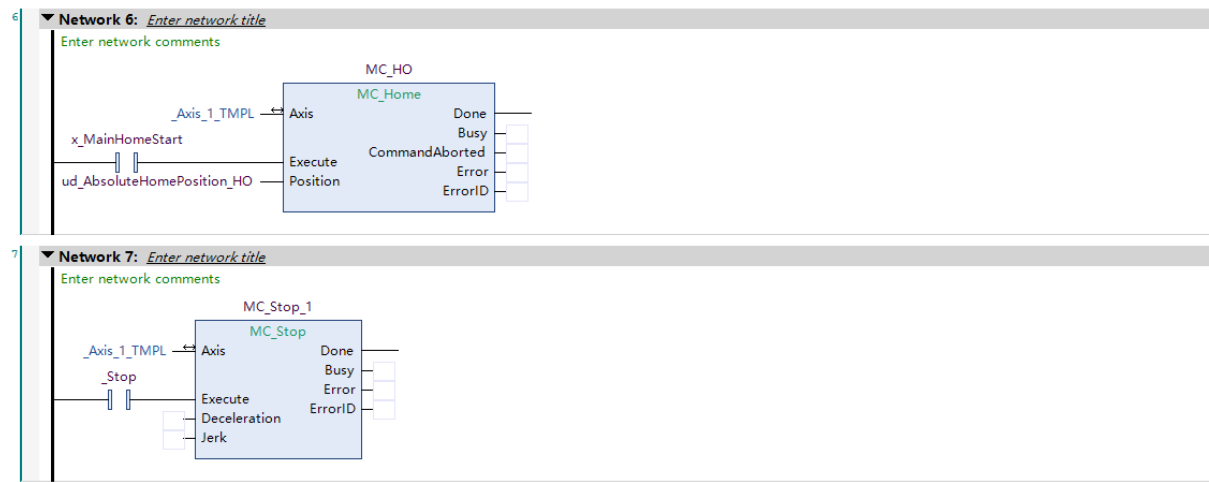


The polypropylene shards are heated and flattened to form thin sheets, which is then passed to the machine that I have programmed by using an HCFA PLC (structured text and ladder logic). The photos of the sheet and the internal circuitry of the machine are provided below.

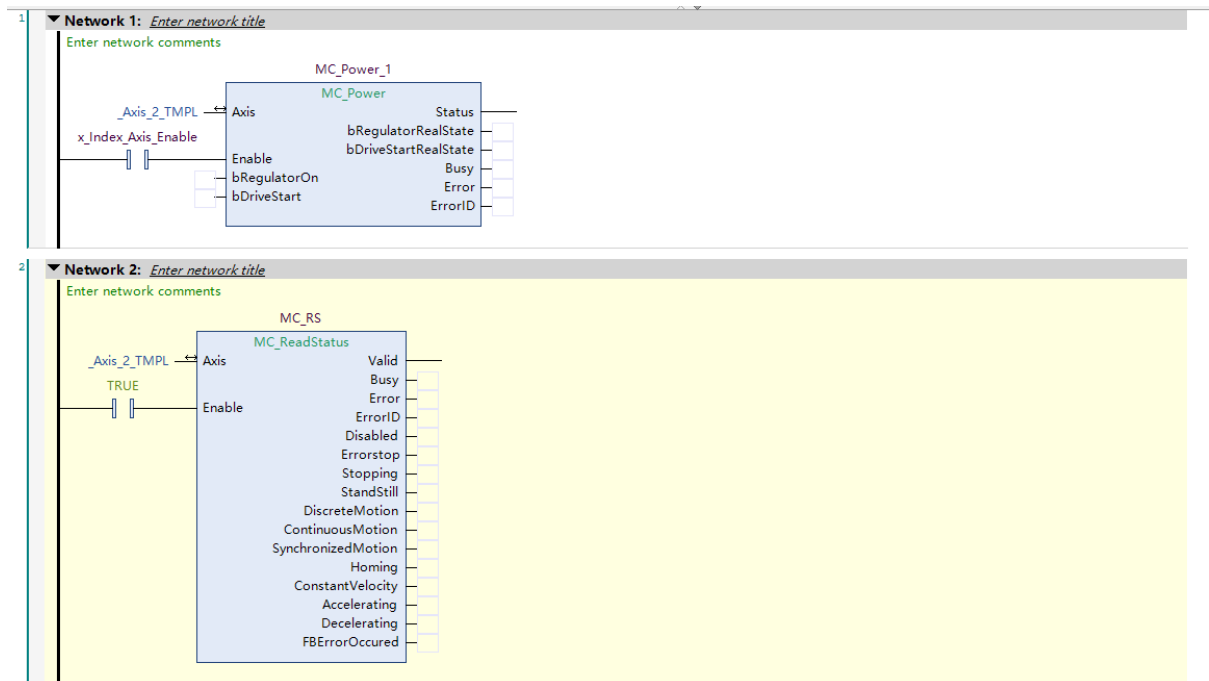


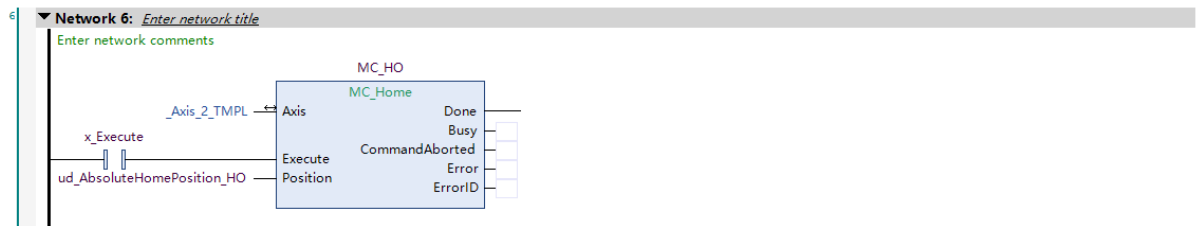
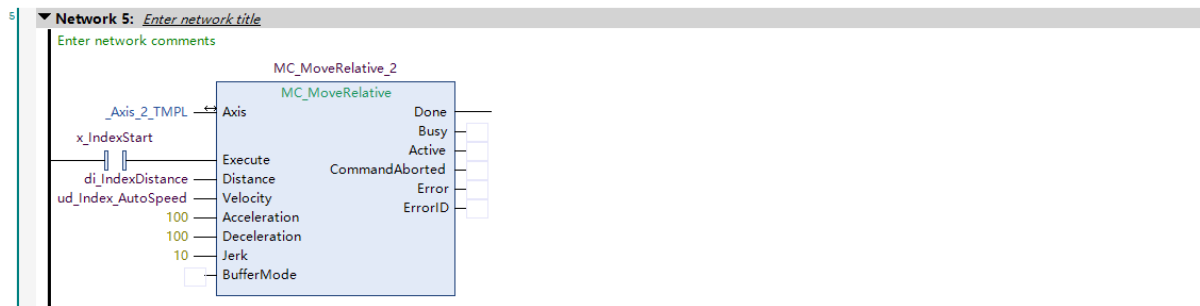
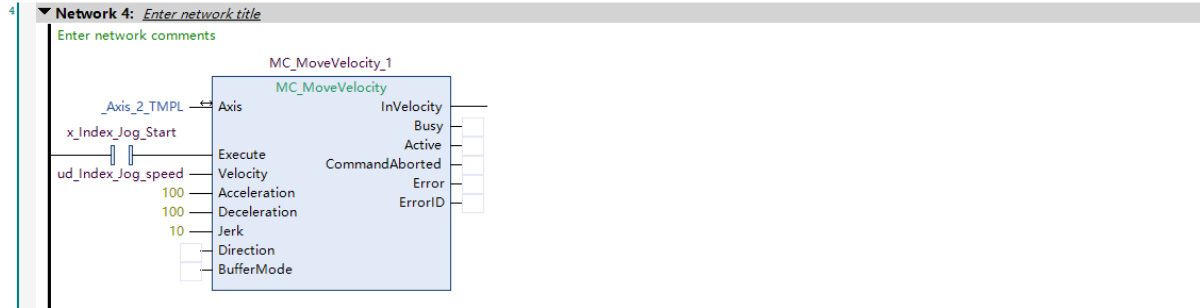
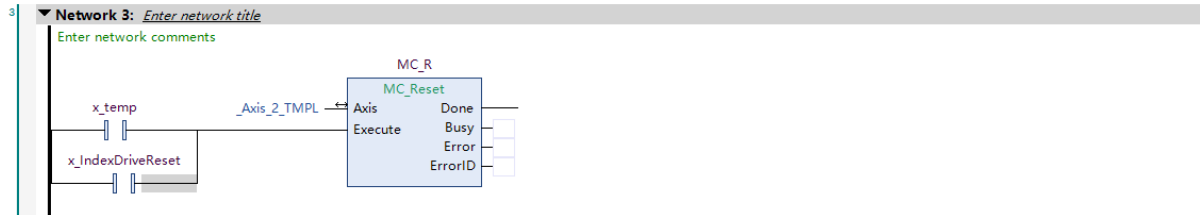
Axis 1 ladder logic:





Axis 2 ladder logic:





The PLC code has produced a result which is that the machine is now fully automated. The machine and the final product of plastic disposable cups are shown below.

